

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Level 1

Lab Manual & Student Workbook

Level: Level 1

Class: Class 6

Activities: 72 Lab Activities + 36 Dormitory Tasks

ICT Teaching Resources Portal

Table of Contents

Chapter 1: Internet and ICT Environment	3
Activity 1.1: Explore Web Browsers	3
Activity 1.2: Google Search Practice	3
Chapter 2: Graphics in ICT	3
Activity 2.1: Drawing with MS Paint	3
Activity 2.2: Creating Symmetric Designs	4
Chapter 3: Audio and Visual Communication	4
Activity 3.1: Recording with Audacity	4
Activity 3.2: Editing Audio Clips	5
Chapter 4: Data Representation and Processing	5
Activity 4.1: Binary Conversion Practice	5
Activity 4.2: Data Analysis Activity	5
Chapter 5: Programming	6
Activity 5.1: Scratch Basics	6
Activity 5.2: Creating Animations	6
Chapter 6: Software Applications	6
Activity 6.1: Word Processing	6
Activity 6.2: Spreadsheet Basics	7
Dormitory Tasks	7
Capstone Projects	9
Computer Club Activities	9

Chapter 1: Internet and ICT Environment

Activity 1.1: Explore Web Browsers

Activity 1.1: Explore Web Browsers

Duration: 40 minutes

Resources Needed:

- Computer with Internet access
- Firefox or Chrome browser
- Printed URL list

Procedure: 1. Locate the browser icon on the desktop or taskbar and double-click to open it. 2. In the address bar, type a URL such as <https://www.wikipedia.org> and press Enter. 3. Observe how the page loads: text, images, and links appear on screen. 4. Click on the search box, type a topic of interest, and press Enter to view results. 5. Use the Back and Forward buttons to navigate between pages, then close the browser.

Questions: 1. What is the difference between the address bar and the search box in a browser? 2. Name two web browsers you know and explain how you identify them. 3. Why is it important to type URLs carefully before pressing Enter?

Activity 1.2: Google Search Practice

Activity 1.2: Google Search Practice

Duration: 40 minutes

Resources Needed:

- Computer with Internet access
- Google search engine
- Notebook for recording results

Procedure: 1. Open Google search at <https://www.google.com> and observe the simple interface. 2. Type a query such as "famous monuments of India" and press Enter. 3. Review the first page of results and note the titles and short descriptions. 4. Refine the search by adding more words, for example "famous monuments of India history". 5. Record three useful results in your notebook and note why each result is helpful.

Questions: 1. How does adding more keywords change the search results you receive? 2. What is the purpose of using quotation marks around a phrase in Google search? 3. How can you tell if a search result is reliable and trustworthy?

Chapter 2: Graphics in ICT

Activity 2.1: Drawing with MS Paint

Activity 2.1: Drawing with MS Paint

Duration: 40 minutes

Resources Needed:

- Computer with MS Paint installed
- Mouse or stylus
- Printed reference images

Procedure: 1. Open MS Paint from the Start menu and explore the toolbar: brushes, shapes, and colors. 2. Select the Rectangle tool and draw a frame on the canvas using a bright color. 3. Use the Pencil tool to sketch a simple house inside the rectangle. 4. Choose the Fill tool to color different parts of your drawing. 5. Save your drawing as a PNG file in your Documents folder.

Questions: 1. What is the difference between the Pencil tool and the Brush tool in MS Paint? 2. How do you undo a mistake you made while drawing? 3. Why is it important to save your work frequently while drawing?

Activity 2.2: Creating Symmetric Designs

Activity 2.2: Creating Symmetric Designs

Duration: 40 minutes

Resources Needed:

- Computer with MS Paint
- Graph paper for planning
- Printed examples of symmetric art

Procedure: 1. Draw a vertical line down the center of the canvas using the Line tool. 2. On the left side of the line, draw a simple shape such as a butterfly wing. 3. Copy and flip the shape horizontally, then place it on the right side. 4. Add colors to both halves, ensuring they match exactly. 5. Compare your design with a partner's work and discuss what makes it symmetric.

Questions: 1. What does it mean for a design to be symmetric? 2. How does flipping an image horizontally help create symmetry? 3. Can you think of examples of symmetry found in nature? Name two.

Chapter 3: Audio and Visual Communication

Activity 3.1: Recording with Audacity

Activity 3.1: Recording with Audacity

Duration: 40 minutes

Resources Needed:

- Computer with Audacity installed
- Microphone or headset
- Printed quick-start guide

Procedure: 1. Open Audacity from the Start menu and identify the Record, Play, and Stop buttons. 2. Click the Record button and speak a short sentence into the microphone. 3. Click Stop to end recording, then Play to listen to your recording. 4. Observe the waveform on the screen and note how loud and soft parts appear. 5. Save your project as an Audacity file and export the recording as an MP3 file.

Questions: 1. What is the difference between recording and playback in Audacity? 2. How can you tell if your recording is too loud or too soft by looking at the waveform? 3. Why is it useful to export your recording as an MP3 file?

Activity 3.2: Editing Audio Clips

Activity 3.2: Editing Audio Clips

Duration: 40 minutes

Resources Needed:

- Computer with Audacity
- Pre-recorded audio clip
- Printed editing shortcuts guide

Procedure: 1. Open an existing audio clip in Audacity and listen to the full recording. 2. Select a small section of the audio that you want to remove and click Delete. 3. Use the Trim tool to keep only the first 10 seconds of the recording. 4. Apply the Amplify effect to make quiet sections louder. 5. Export the edited audio as a new MP3 file and compare it with the original.

Questions: 1. What is the difference between cutting and deleting audio in Audacity? 2. How does the Amplify effect change the way your audio sounds? 3. When might you need to edit an audio clip before sharing it with others?

Chapter 4: Data Representation and Processing

Activity 4.1: Binary Conversion Practice

Activity 4.1: Binary Conversion Practice

Duration: 40 minutes

Resources Needed:

- Printed binary conversion chart
- Pencil and paper
- Calculator (optional)

Procedure: 1. Study the place values in binary: 128, 64, 32, 16, 8, 4, 2, 1. 2. Convert the decimal number 13 to binary by subtracting place values. 3. Convert the decimal number 25 to binary using the same method. 4. Convert the binary number 1011 back to decimal and check your answer. 5. Create a table comparing at least five decimal and binary number pairs.

Questions: 1. How many digits does the binary number system use? 2. What is the place value of the fourth position from the right in binary? 3. Why is the binary number system important for computers?

Activity 4.2: Data Analysis Activity

Activity 4.2: Data Analysis Activity

Duration: 40 minutes

Resources Needed:

- Printed survey data sheet
- Pencil and ruler
- Graph paper

Procedure: 1. Read the provided survey data about favorite fruits in your class. 2. Tally the responses for each fruit and record the total counts. 3. Create a bar graph on graph paper to represent the survey results. 4. Write two observations about the data based on your bar graph. 5. Share your graph with a classmate and compare your observations.

Questions: 1. What type of graph is best for comparing the popularity of different items? 2. How does organizing data into categories help in understanding information? 3. What would happen if you did not label the axes on your bar graph?

Chapter 5: Programming

Activity 5.1: Scratch Basics

Activity 5.1: Scratch Basics

Duration: 40 minutes

Resources Needed:

- Computer with Scratch installed or internet access
- Printed Scratch block reference
- Notebook for planning

Procedure: 1. Open Scratch and explore the interface: sprite, stage, and block palette. 2. Select a sprite and add a "say" block to make it display a message. 3. Add a "move" block to make the sprite move forward on the stage. 4. Use a "repeat" block to make the movement happen multiple times. 5. Click the green flag to run your program and observe the sprite's behavior.

Questions: 1. What is a sprite in Scratch and how do you add one to your project? 2. How does changing the number inside a repeat block affect the program? 3. What is the difference between clicking a block directly and using the green flag?

Activity 5.2: Creating Animations

Activity 5.2: Creating Animations

Duration: 40 minutes

Resources Needed:

- Computer with Scratch
- Printed animation planning worksheet
- Story outline

Procedure: 1. Plan a short animation story with a beginning, middle, and end. 2. Create two sprites and give each a unique costume. 3. Use the "switch costume" block to make a sprite appear to walk. 4. Add a backdrop change to show a change in location. 5. Test your animation, fix any issues, and present it to the class.

Questions: 1. What is the difference between a sprite and a backdrop in Scratch? 2. How can you control the speed of an animation in Scratch? 3. Why is planning your story important before you start coding?

Chapter 6: Software Applications

Activity 6.1: Word Processing

Activity 6.1: Word Processing

Duration: 40 minutes

Resources Needed:

- Computer with LibreOffice Writer or Microsoft Word
- Printed formatting guide

- Sample text document

Procedure: 1. Open the word processor and create a new blank document. 2. Type a short paragraph about your favorite hobby. 3. Select the title text and apply bold, underline, and a larger font size. 4. Change the font color and alignment of the paragraph to center. 5. Save the document as a DOCX file in your Documents folder.

Questions: 1. What are the advantages of using a word processor over handwriting? 2. How do you select text in a word processor? 3. What file format would you use to share a document with someone who does not have the same software?

Activity 6.2: Spreadsheet Basics

Activity 6.2: Spreadsheet Basics

Duration: 40 minutes

Resources Needed:

- Computer with LibreOffice Calc or Microsoft Excel
- Printed formula reference
- Sample data sheet

Procedure: 1. Open the spreadsheet program and explore the grid of cells and columns. 2. Enter the names of five subjects in column A and marks in column B. 3. In column C, type a formula to calculate the average of the marks. 4. Use the SUM function to find the total of all marks in a separate cell. 5. Save your spreadsheet and print a copy showing the formulas.

Questions: 1. What is the difference between a cell and a column in a spreadsheet? 2. How do you enter a formula that starts with an equals sign in a spreadsheet? 3. Why are spreadsheets useful for organizing and calculating data?

Dormitory Tasks

Task 1: Browser Features Identification

Duration: 30 minutes

Draw a labeled diagram of a web browser window. Identify the address bar, back button, forward button, refresh button, and tab bar. Write one sentence explaining the purpose of each feature.

Task 2: Search Strategy Planning

Duration: 30 minutes

Write a list of five effective search queries for finding information about the water cycle. Explain why each query would give better results than a generic search like "water".

Task 3: Drawing Tool Comparison

Duration: 30 minutes

Create a table comparing the Pencil, Brush, and Shape tools in MS Paint. For each tool, describe its function, one use case, and one limitation.

Task 4: Symmetry Art Planning

Duration: 30 minutes

On graph paper, plan a symmetric design by drawing only the left half. Leave the right half blank and describe how you would flip it to complete the design.

Task 5: Audio Recording Checklist

Duration: 30 minutes

Create a checklist of five things to do before, during, and after recording audio in Audacity. Include tips for reducing background noise and ensuring clear sound.

Task 6: Audio Editing Steps

Duration: 30 minutes

Write step-by-step instructions for trimming an audio clip to keep only the first 15 seconds in Audacity. Include keyboard shortcuts where possible.

Task 7: Binary Number Worksheet

Duration: 30 minutes

Convert the following decimal numbers to binary: 7, 10, 18, 30, 42. Show your working for each conversion using the place value method.

Task 8: Data Collection Plan

Duration: 30 minutes

Design a survey to collect data about the modes of transport used by students in your class. Create the questions, categories, and a tally chart template.

Task 9: Scratch Block Matching

Duration: 30 minutes

Match each Scratch block to its function: move, say, repeat, switch costume, when green flag clicked. Draw or describe a simple program using all five blocks.

Task 10: Animation Storyboard

Duration: 30 minutes

Create a storyboard with four frames for a short Scratch animation. Include the sprites, backdrops, and actions for each frame.

Task 11: Word Processing Formatting Guide

Duration: 30 minutes

Write a reference guide listing five text formatting options in a word processor. For each option, describe how to apply it and when to use it.

Task 12: Spreadsheet Formula Practice

Duration: 30 minutes

Write the correct spreadsheet formulas for the following tasks: sum of cells A1 to A5, average of cells B1 to B4, largest value in cells C1 to C6, and count of cells D1 to D3 that contain numbers.

Capstone Projects

Task C1: Digital Research Portfolio

Duration: 2 hours

Choose a topic related to Indian culture or science. Use a web browser to find at least five reliable sources. Create a word processing document that includes a title, introduction, three main points with evidence from your sources, a conclusion, and a bibliography. Present your findings to the class using at least three formatted elements such as headings, bullet points, and bold text.

Task C2: Interactive Scratch Story

Duration: 2 hours

Design and create a Scratch project that tells an interactive story with at least three scenes. The project must include at least two sprites with different costumes, backdrop changes, user interaction through keyboard or mouse, and a sound or speech element. Write a planning document before coding and reflect on what you learned after completing the project.

Computer Club Activities

Task A1: Internet Safety Poster

Duration: 45 minutes

Research five rules for staying safe online. Create a visually appealing poster using MS Paint or a word processor that presents each rule with an illustration or icon. Share your poster with classmates and discuss why each rule is important.

Task A2: Pixel Art Challenge

Duration: 45 minutes

Using graph paper, design a pixel art character or object. Each square represents one pixel. Use at least four colors and create a design that is at least 10 by 10 squares. Recreate your design digitally in MS Paint.

Task A3: Podcast Script Writing

Duration: 45 minutes

Write a three-minute podcast script about a topic of your choice. Include an introduction, three discussion points, and a conclusion. Practice reading it aloud with appropriate pauses and emphasis.

Task A4: Number System Quiz

Duration: 45 minutes

Create a quiz with ten questions about the binary and decimal number systems. Include multiple choice, true or false, and conversion questions. Provide an answer key for your quiz and administer it to a classmate.

Task A5: Scratch Game Design

Duration: 45 minutes

Design a simple game in Scratch where a sprite collects falling objects. The game should include score tracking, a timer, and a win or lose condition. Write a design document explaining your game mechanics before you start coding.

Task A6: Typing Speed Challenge

Duration: 45 minutes

Practice typing using an online typing tutor or a typing game. Record your words per minute and accuracy after three attempts. Create a short report comparing your results and setting goals for improvement.